

STA 35C: Statistical Data Science III

Lecture 6: Multiple Linear Regression & Polynomial Regression

Dogyoon Song

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Recap: Simple linear regression

- Linear model: $Y = \beta_0 + \beta_1 X + \varepsilon$
- *Least squares estimation* chooses $(\hat{\beta}_0, \hat{\beta}_1)$ minimizing

$$\text{RSS}(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

- *Interpretation:* β_1 is the average change in the mean response per one-unit increase in X
 - Associational, not necessarily causation
- *Inference:* $\hat{\beta}_0$ and $\hat{\beta}_1$ are random variables
 - Standard errors, confidence intervals, and tests
 - Rejecting $H_0 : \beta_1 = 0$ gives evidence of a nonzero slope
- Model fit is summarized by RSE and

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad \text{where} \quad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

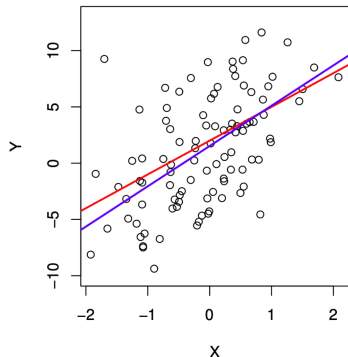


Figure: Least squares coefficient estimates from 100 data points.
Red: population regression line,
Blue: least squares line
[JWHT21, Figure 3.3].

Completing Lecture 5: The R^2 statistic

The mean-only prediction is $\bar{y}$; the fitted (linear) regression model predicts $\hat{y}_i$.

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

- R^2 is the proportion of observed variation captured by the fitted model
- For least squares with an intercept and $\text{TSS} > 0$, training $R^2 \in [0, 1]$ and is scale-free
- High training R^2 does not guarantee a significant slope, accurate individual predictions, or good test performance

Example (Four observations)

Data (x_i, y_i) : $(1, 1), (1, 3), (2, 3), (2, 5)$, so $\bar{y} = 3$

Least squares: $\hat{y} = 2x$, giving fitted values 2, 2, 4, 4

$\text{TSS} = 4 + 0 + 0 + 4 = 8$, $\text{RSS} = 1 + 1 + 1 + 1 = 4$

$R^2 = 1 - \frac{4}{8} = \frac{1}{2}$: the line captures 50% of the observed variation

Agenda

Last time: Simple linear regression

- Model, least squares, inference, and model fit

Today: Extending simple linear regression

- Multiple linear regression
 - Motivation: *"What if we have more than one predictor?"* $X_1, X_2, \dots$
 - Model
 - Estimation
 - Inference
 - Model fit
 - Prediction intervals
- Polynomial regression
 - Motivation: *"What if X_1 and X_2 interact, or if Y depends on X^2 instead of X ?"*

Motivation for multiple linear regression

Recall the **Advertising** dataset and the three separate simple linear regression lines:

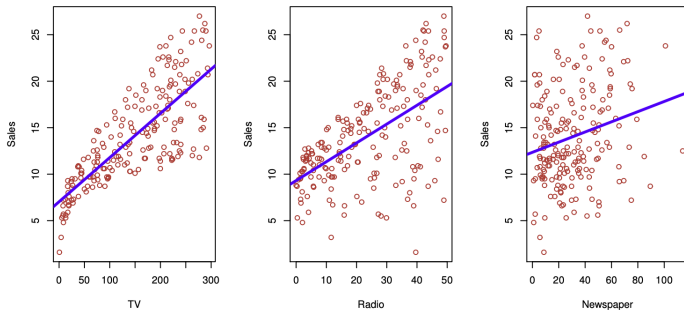


Figure: The **Advertising** data set: **Sales** of a product in 200 different markets against advertising budgets for three media: **TV**, **Radio**, and **Newspaper** [JWHT21, Figure 2.1].

Problem: Each simple linear regression line ignores the other two predictors

Question: Can we extend our analysis to accommodate *all* predictors simultaneously?

Multiple linear regression: Setup

We predict Y using multiple variables $X_1, X_2, \dots, X_p$, assuming:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon,$$

- *Model parameters:* $\beta_0, \beta_1, \dots, \beta_p$ are fixed, unknown constants
- ε is an error term, satisfying $\mathbb{E}[\varepsilon \mid X_1, \dots, X_p] = 0$

The coefficient β_j is interpreted as the **average change** in the **mean response** for a one-unit increase in X_j , **holding all other predictors fixed**

- This is an associational interpretation, not necessarily a causal one

Once we *estimate* $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ from training data, we can predict

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

Visualizing multiple linear regression

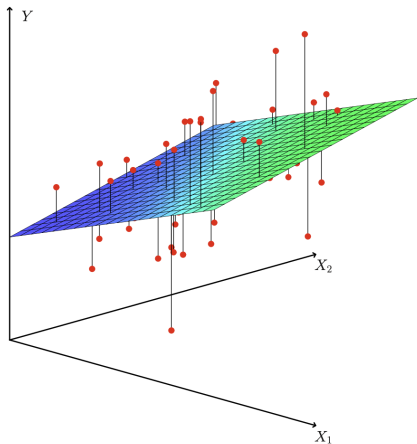


Figure: An illustration of multiple linear regression [JWHT21, Figure 3.4]. With two predictors, the fitted surface is a plane; with many predictors, it is a hyperplane.

Coefficient estimation via least squares

Coefficients $\beta_0, \beta_1, \dots, \beta_p$ must be estimated from data $(x_1, y_1), \dots, (x_n, y_n)$, where

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip}).$$

Again, we use the **least squares** criterion:

- The *least squares* approach chooses $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ that minimize the RSS:

$$\text{RSS}(b_0, \dots, b_p) = \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip})^2$$

- (Optional) In matrix form¹, when $\mathbf{X}^\top \mathbf{X}$ is invertible:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad \text{where} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

¹This can be derived by setting the partial derivatives of RSS to 0, but is beyond the scope of STA 35C

Pop-up quiz: Interpreting coefficients

Scenario: We fit a multiple linear regression model on the **Advertising** dataset:

$$\text{Sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \varepsilon.$$

Suppose that we obtain: $\hat{\beta}_1 = 0.04$, $\hat{\beta}_2 = 0.18$, $\hat{\beta}_3 = -0.02$.

Question: Which statement *best* describes the meaning of $\hat{\beta}_1 = 0.04$ in this model?

Multiple-choice answers:

- A) TV advertising alone explains 4% of the variation in **Sales**
- B) A 1-unit increase in **TV** causes **Sales** to increase by exactly 0.04
- C) Holding **Radio** and **Newspaper** fixed, a 1-unit increase in **TV** is associated with an average increase of 0.04 units in predicted **Sales**
- D) The effect of **TV** is 0.04 only when **Radio** and **Newspaper** are both 0

Answer: C.

In multiple regression, a coefficient measures the average change in the predicted response for a one-unit increase in that predictor, holding the others fixed

Some key questions with multiple predictors

When we perform multiple linear regression, we often want to answer questions like:

- Are predictors $X_1, \dots, X_p$ related to Y (i.e., do they help predict Y)?
- Which subset of $X_1, \dots, X_p$ is most important?
- How well does the model fit the data?
- Given new predictor values, what response value should we predict and how accurate is that prediction?

Let's address these questions one by one

Hypothesis testing for relationship between Y and each X_j

Recall from simple linear regression:

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 \neq 0, \quad t = \frac{\hat{\beta}_1}{\widehat{\text{SE}}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0.$$

- **Idea:** If H_0 is true, t follows the t-distribution and will likely be close to 0
- Thus, a very large $|t|$ value is an evidence against $H_0 \rightarrow$ reject H_0

In multiple linear regression, we do the same, but for each predictor X_j :

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0, \quad t = \frac{\hat{\beta}_j}{\widehat{\text{SE}}(\hat{\beta}_j)} \sim t_{n-p-1} \text{ under } H_0.$$

- Large $|t_j|$ is evidence against H_0 ; use the critical value for the chosen significance level
- This tests a zero partial slope (Y vs X_j) *after accounting for the other predictors*
- The answer can differ from simple regression setting when predictors are correlated
- Also, we may instead ask whether *any* predictor is useful at all (will see shortly)

Advertising example: Simple vs. multiple regressions

Q: Is **newspaper** associated with **sales**, before and after adjusting for other media?

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

	Coefficient	Std. error	t-statistic	p-value
Intercept	9.312	0.563	16.54	< 0.0001
radio	0.203	0.020	9.92	< 0.0001

	Coefficient	Std. error	t-statistic	p-value
Intercept	12.351	0.621	19.88	< 0.0001
newspaper	0.055	0.017	3.30	0.00115

Figure: Separate simple regressions suggest **TV**, **radio**, and **newspaper** are all significant [JWHT21, Tables 3.1 & 3.3].

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Figure: Multiple regression suggests **newspaper** is not significant [JWHT21, Table 3.4].

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Figure: Correlation matrix for **TV**, **radio**, **newspaper**, and **sales** [JWHT21, Table 3.5].

Takeaway: In multiple regression, β_j measures the association between X_j and Y *holding the other predictors fixed*; correlated predictors can therefore change significance

Advertising example: Testing single vs. any predictor

Q: Is “any” (=at least one) of TV, radio, newspaper useful in predicting sales?

We test the *joint* hypothesis:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0$$

At level α , use the *F-statistic* and its $(1 - \alpha)$ quantile:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)} \quad \begin{cases} \text{Reject } H_0 & \text{if } F > F_{p, n-p-1; 1-\alpha}, \\ \text{Do not reject } H_0 & \text{otherwise.} \end{cases}$$

Rationale: If H_0 is true, both mean squares in the ratio estimate σ^2

- If all predictors are useless, adding them should not reduce RSS much, so F should be near 1
- A large F means the RSS reduction is larger than expected from noise alone
- Under H_0 , F follows an F -distribution with $(p, n - p - 1)$ degrees of freedom

We will discuss how to select “important” predictors later in this course (after Midterm 1)

Evaluating the model fit

Training fit of a multiple linear regression² can be measured by the RSE or the R^2

- **Residual standard error (RSE):** “typical size of deviation, in the same units as Y ”

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n - p - 1}} \quad \text{where} \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **The R^2 :** “the proportion of observed variation captured by the fitted model”

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{where} \quad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

- On the *same observations*, adding predictors cannot reduce training R^2
- “**Adjusted**” R^2 compensates for adding predictors:

$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)}$$

- It rises only when the added predictors improve fit to justify the extra complexity

²Full-rank model, $n > p + 1$, $\text{TSS} > 0$

Pop-up quiz: R^2 vs. adjusted R^2

Scenario: With $n = 22$ observations, suppose a one-predictor model with an intercept has

$$R^2 = 0.80, \quad R^2_{\text{adj}} = 0.79.$$

After adding a second predictor X_2 (suspected to be mostly noise), we get:

$$R^2 = 0.805, \quad R^2_{\text{adj}} \approx 0.784.$$

Question: Why did R^2 go up while R^2_{adj} went down?

- A) There must be a calculation error; if R^2 increases, R^2_{adj} must also increase.
- B) X_2 reduced RSS, but not enough to offset losing one residual degree of freedom; adjusted R^2 decreased
- C) Adjusted R^2 always decreases whenever we add a predictor, no matter how useful it is
- D) R^2 does not measure model fit at all, whereas R^2_{adj} is the only valid measure of fit

Answer: B.

Here, $1 - (1 - 0.805) \frac{21}{19} \approx 0.784 < 0.79$. This is a training-fit comparison, not proof that X_2 contains no signal

Confidence intervals and prediction intervals

With $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$, we predict $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$

Under the stated model assumptions, consider two kinds of uncertainty:

- $\hat{y} = \hat{f}_{\hat{\beta}}(x)$ only *estimates* $f_{\beta}(x) = \beta_0 + \sum_{j=1}^p \beta_j x_j$ → variability due to estimates $\hat{\beta}_j$
- $y = f(x) + \varepsilon$ has an error term → additional variability due to noise

Confidence interval for the mean response $f(x)$:

- Reflects uncertainty in prediction due to estimating coefficients
- Across repeated samples, 95% of CIs would contain the true *mean response* at x

Prediction interval for a new response $Y_{\text{new}} \mid X = x$:

- Accounts for uncertainty in both $\hat{y} = \hat{f}(x)$ *and* the random noise ε
- Over repeated training samples and new observations at fixed x , 95% of PIs cover Y_{new}
- For the same fit, x , and confidence level, a PI is wider than a mean-response CI

In **R** (`x0`: a data frame with matching predictor names):

```
predict(model, newdata = x0, interval = "confidence", level = 0.95)
```

```
predict(model, newdata = x0, interval = "prediction", level = 0.95)
```

What if there is a non-linear relationship?

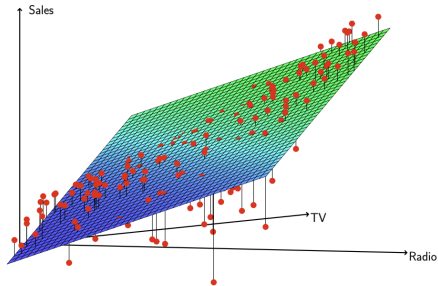


Figure: Pronounced synergy between **TV** and **Radio**; positive residuals cluster along the 45-degree line [JWHT21, Figure 3.5].

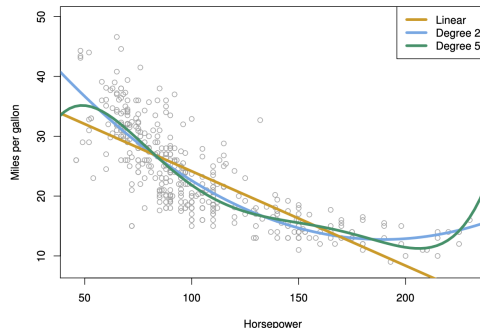


Figure: A non-linear relationship between **mpg** and **horsepower** is noticeable [JWHT21, Figure 3.8].

→ We can enrich the model by adding **interaction** terms (**TV** $\times$ **Radio**) or **non-linear** terms (**horsepower**²) to capture these effects

Polynomial regression: Higher-order and interaction terms

Polynomial regression extends the linear model by including powers of predictors³:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_d X^d + \varepsilon$$

- Treated as multiple linear regression on transformed predictors $(X, X^2, \dots, X^d)$
- Although non-linear in X , the model is still linear in the coefficients β_j

Example: Interaction effect (synergy between **TV** and **Radio**)

$$\begin{aligned}\text{Sales} &= \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \underbrace{\text{TV} \times \text{Radio}}_{\text{interaction term}} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \text{Radio}) \text{TV} + \beta_2 \text{Radio} + \varepsilon\end{aligned}$$

- Holding **Radio** fixed, the TV slope for mean sales is $\beta_1 + \beta_3 \text{Radio}$, not β_1 alone

Example: Quadratic model

$$\text{mpg} = \beta_0 + \beta_1 \text{horsepower} + \beta_2 \underbrace{\text{horsepower}^2}_{\text{nonlinear term}} + \varepsilon$$

³More generally, $Y = \sum_{\alpha: |\alpha| \leq d} \beta_{\alpha} \mathbf{X}^{\alpha} + \varepsilon$ where $\alpha = (\alpha_1, \dots, \alpha_p)$ and $\mathbf{X}^{\alpha} = X_1^{\alpha_1} X_2^{\alpha_2} \dots X_p^{\alpha_p}$

Model flexibility and the risk of overfitting

A more flexible model can fit the training data more closely

- For least squares on the same observations, adding terms cannot increase training RSS
- Training RSS = RSS measured on the training data

But a better fit on the training data does not necessarily mean a better model

- Our actual goal is good performance on new, unseen data
- A model that fits the training set extremely well may perform worse on a test set

Overfitting: If a model is too flexible, it can fit noise as well as signal

- Typical symptom: very small training error, but noticeably larger test error
- Compare training fit with performance on new data, not with training fit alone
- We will return to this issue repeatedly throughout the course

Wrap-up

Multiple linear regression extends simple linear regression to several predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

- Each slope coefficient β_j describes the change in the mean response per unit increase in X_j , holding the other predictors fixed
- Individual t -tests and the overall F -test answer different questions: Association to one predictor vs. any predictor
- Model fit is summarized by RSE, R^2 , and adjusted R^2 ; adjusted R^2 helps compare models with different numbers of predictors
- Mean-response confidence intervals target $f(x)$; prediction intervals target a new observation
- Interaction and polynomial terms make the model more flexible, but extra flexibility increases the risk of overfitting

Suggested reading: [JWHT21, Secs. 3.2, 3.3.2; optional Sec. 7.1]

Next lecture: Qualitative predictors and pitfalls in linear regression

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.